

Samuel Handel

• Phone: 757-575-5637 • Email: shand001@odu.edu

Education

Old Dominion University, Norfolk, Virginia

M.S. in Data Science • Concentration: Artificial Intelligence and Machine Learning

Started Fall 2026 | Expected Graduation: 2028

Virginia Tech, Blacksburg, Virginia

B.S. in Environmental Data Science • Minor: Geographic Information Systems (GIS) Overall GPA: 3.33 | Major GPA: 3.58

Relevant Coursework: Environmental Data Science, Systems Analysis, Database Management and Design, Modeling with GIS, Data Matter.

Technical Skills

Programming & Analysis: Python, R, SQL, JMP, VBA, Excel

GIS & Mapping Tools: ArcGIS Pro, Google Earth Engine

Data Visualization: R (Tidyverse, ggplot2, dplyr, tidyr, base R)

Other: Database Management, Data Cleaning, Automated Machine Learning, Multidimensional Scaling (MDS), Text Analysis, Dimension Reduction, Linear Regression, Spatial Autocorrelation

Leadership & Involvement

Geography Club — Past Member (Aug 2025 – May 2026)

Alpha Epsilon Pi, Virginia Tech — Alumni (Nov 2023 – May 2026)

Minor Board Member – Logistics Chair (Jan 2024 – May 2024); Community Engagement Chair (Jan 2025 – May 2025)

-Managed chapter logistics and engagement initiatives for 50+ members

AI Club, Old Dominion University — Member (Sep 2026 – Present)

Memberships

Student Member, American Statistical Association

Student Member, Virginia Public Health Association

Publications & Presentations

County-Level Mental Health Burden and Behavioral Health Workforce Access in Virginia: An Ecological Analysis — Virginia Public Health Association (VPHA) Conference, Sep 24, 2026 (3rd Author)

- Conducted a cross-sectional ecological analysis across all 133 Virginia counties and independent cities examining the association between behavioral health workforce availability and county-level mental health burden
- Integrated data from the U.S. Census Bureau, HRSA Area Health Resources Files, and CDC PLACES; applied multiple linear regression with stepwise AIC selection (adjusted $R^2 = 0.88$)
- Applied K-means clustering on MDS-reduced data to identify four distinct county groupings by geographic and socioeconomic profile

Projects

NEON Biomass ML Model – Spring 2024

- Built a predictive workflow using Tidyverse and H2O AutoML on a dataset of 1000+ carbon storage records in vegetation
- Evaluated the model performance with RMSE and R^2

P-Hacking Evaluation – Fall 2025

- Converted self-reported health data to a binary format and applied Gower Distance
- Used Multidimensional Scaling, K-means Clustering, and Linear Regression to evaluate statistical biases of P-Hacking

HBEF Watershed 3 TOPMODEL Explorer – Capstone - Spring 2026

- Built an interactive R Shiny dashboard implementing TOPMODEL for Watershed 3 of the Hubbard Brook Experimental Forest, a catchment known for being difficult to model due to complex terrain and subsurface flow
- Optimized parameters via Nelder–Mead, achieving NSE averaging ~ 0.4 across a 50-year record (up to ~ 0.7 in best-performing years), and analyzed interannual variability in model fit
- Used sf/raster for spatial processing (topographic wetness index, dynamic saturation mapping) and built interactive visualizations comparing simulated vs. observed discharge